Software Requirements Specification

Introduction to Software Engineering (15031001) · Week 3 · Requirements & Specification

Field	This example
Team name	16RideRight
Project title	MFU Housing Translator — An Off-Campus Housing Assistant for International Students
SDLC chosen in M1 + reason	**Incremental** — build and test the core housing translation and information extraction features first, then add comparison, distance, and Thai reply generation features based on user feedback.
Date / version	-
GitHub repo	-https://github.com/Kennethkaungsatt/BaanRight-mfu-housing-demo-

1. Introduction

1.1 Purpose and scope

International students at Mae Fah Luang University (MFU) are only guaranteed campus housing for their first year. After that, they need to find off-campus accommodation, but housing information is difficult to find because listings are scattered across different Facebook groups and are often written only in Thai. Students may also have difficulty identifying important information such as price, deposit, availability, location, and contact details.

MFU Housing Translator is a Chrome extension designed to help international students understand and compare Thai housing listings. Students can paste the text of a housing post. The system translates the listing into English, extracts important housing information, helps students understand the approximate distance from MFU, allows them to save and compare listings, and generates Thai reply messages that students can send to landlords.

The initial MVP focuses on processing individual housing listings and presenting the key information clearly. The system does not directly search or scrape Facebook groups.

1.2 Users and stakeholders

Name / role	Kind	What they need	What they fear / forbid
International Students (primary user)	primary user	Understand Thai housing listings and compare accommodation options	Wrongly translated or extracted information leading to a bad housing decision
Landlord / housing provider	secondary stakeholder	Receive inquiries from international students	Incorrect information or inappropriate messages being sent
Development team	project team	Clear, testable requirements that can be implemented within the course timeframe	Requirements becoming too large or unrealistic for the project

2. Overall description

2.1 Product context — what it does / does not

IN SCOPE (MVP this term)	OUT OF SCOPE (explicit promise)
FR-1 Input a housing listing **FR-2** Translate Thai housing Listing **FR-3** Extract and Save key housing information **FR-4** Generate common Thai messages to landlords **FR-5** Translate selected text directly from webpages	- Automatically scraping Facebook posts. - Booking or reserving accommodation. - Making rental payments. - Automatically contacting landlords. - Scam or red-flag detection. - Guaranteeing that a housing listing is legitimate. - Guaranteeing the accuracy of price or availability. - Providing legal advice about rental contracts. - Determining whether a rental contract is fair or unfair. - Managing rental contracts. -Shared housing database between students. User login/account system. -Building a complete housing marketplace. -Screenshot OCR in the current MVP.

2.2 Assumptions and constraints

- Assumptions: Students can access housing posts on Facebook or other webpages.
- Assumptions: Students can copy housing-post text into BaanRight.
- Assumptions: Housing posts contain at least some information that can be extracted.
- Assumptions: Students will verify important housing information with the landlord before making a final decision.
- Assumptions: Students understand that automatically extracted information may contain errors.

- Constraints: BaanRight is implemented as a **Chrome extension**.
- Constraints: The system does not directly access or scrape Facebook groups.
- Constraints: Translation and information extraction may contain errors and must be checked by the student.
- Constraints: The prototype has a limited development period.

3. Functional requirements

FR-1 — Input a Housing Listing

User story: As an international student, I want to paste Thai housing-post text, so that I can process the listing without manually retyping the information.
Pain this trace to: Housing information is contained inside long Facebook posts, and students need to manually copy or organize the information before they can understand or compare different listings.

1. Given the student pastes housing-post text, when they submit it, then the system accepts and displays the submitted text.

FR-2 — Translate Thai Housing Listing

User story: As an international student, I want Thai housing information translated into English, so that I can understand the listing without translating it manually.
Pain this trace to: Many housing listings in MFU-related Facebook groups are written mainly in Thai, making it difficult for international students who cannot read Thai well to understand the available accommodation.

1. Given a Thai housing listing is submitted, when translation is requested, then the system displays an English translation.
2. Given the translation is generated, when the student views the result, then the original Thai text remains available for comparison.

FR-3 — Extract and Save Key housing information

User story: As an MFU international student, I want BaanRight to extract important information such as price, deposit, location, availability, and phone number, so that I can quickly understand and compare housing listings later.
Pain this traces to: Important housing information is often mixed into long Thai posts instead of being presented in a consistent format, so students have difficulty finding and comparing basic details.

1. Given a housing post contains a monthly rental price, when the listing is processed, then the extracted price is displayed separately.
2. Given a housing post contains a phone number, when the listing is processed, then the phone number is displayed separately.
3. Given a housing post contains a deposit, when the listing is processed, then the deposit information is displayed separately.
4. Given a housing post contains a location,when the listing is processed,then the extracted location is displayed separately.
5. Given a housing post contains availability information,when the listings is processed ,then the availability information is displayed separately.
6. Given a listing has been processed,when the student chooses to save the listing,then the extracted housing information is stored in the student's saved listing.
7. Given the student has saved multiple listing,when the student opens their saved listing,then the system displays the saved housing information so that the student can compare the options.
8. Given a housing post does not contain one of the requested fields, when the listing is processed, then BaanRight indicates that the information was not found instead of inventing a value.

Current prototype: ✅ Working
Implementation: Rule-based extraction, not AI-based extraction.

FR-4 — Generate Thai reply to landlord

User story: As a MFU international student, I want to BaanRight to generate common Thai question for a landlord,so that i can communicate with Thai landlord without manually translating each question.

Pain this traces to: international students may understand a housing listing after translation but still struggle to communicate with Thai-speaking landlords.

1. Given the student selects a common question such as availability or deposit, when they request a reply, then BaanRight generates the corresponding Thai message
2. Given a Thai reply has been generated, when the student selects copy, then the Thai message is copied so that it can be sent to the landlord.
3. Given the student reviews the generated message, when they decide not to send it, then BaanRight does not automatically contact the landlord.

Current prototype: ✅ Working
Current limitation: The reply generator currently provides five fixed questions rather than fully cust.

FR-5 — Translate Selected Text from a Webpage

User story: As an MFU international student,I want to translate selected Thai text directly from a webpage,so that I can process housing information without repeatedly copying and pasting text.

Pain this traces to: Students browsing Facebook housing groups have to manually copy text from posts before using a translation tool, which adds extra steps.

1. Given the student selects Thai text on a webpage , when they choose BaanRight's translate option, then the selected text is sent to BaanRight for translation.
2. Given the student right-clicks selected text, when they choose "translate with BaanRight",then the selected text is processed by BaanRight.

Current prototype : ✅ Working

4. Non-functional requirements

ID	Kind	Statement (testable)	How we will check
NFR-1	performance	BaanRight shall display the translation result within **10 seconds** for a normal text housing listing under normal network conditions.	Stopwatch over 3 trials from our group members
NFR-2	performance	BaanRight shall display extracted housing fields within **5 seconds** after translation/input processing is completed.	Stopwatch over 3 trials from our group members
NFR-3	usability	A first-time student shall be able to paste a housing listing and start processing it within 30 seconds without assistance.	Timed test with a student unfamiliar with BaanRight
NFR-4	reliability	When the translation request fails, BaanRight shall display an error message rather than showing an empty or misleading translation result	Disconnect the device from the internet, submit a Thai housing listing, and observe whether BaanRight displays an errormessage.(test result-passed✅)When tested without an internet connection, BaanRight displayed **"Couldn't translate that: Failed to fetch"** instead of displaying a misleading translation
NFR-5	security / privacy (required)	Saved housing listings shall remain available only in the browser storage of the student who saved them. (baanright current prototype uses - chrome.storage.local)	1.Browser A → save Listing A. 2.Close BaanRight. 3.Open Browser B / another browser profile. 4.Open BaanRight. 5.Check saved listings. passed✅

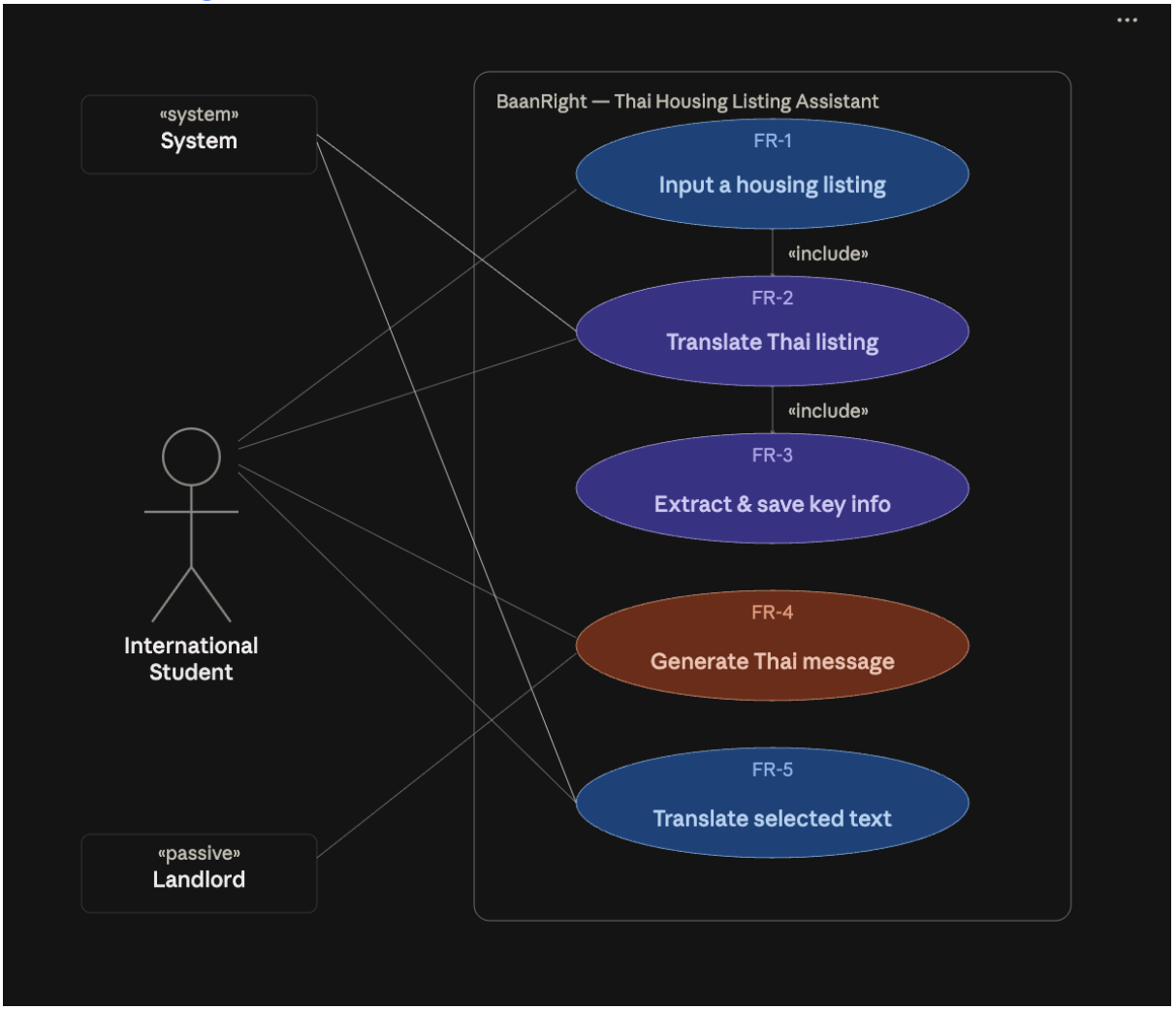
5. Use cases

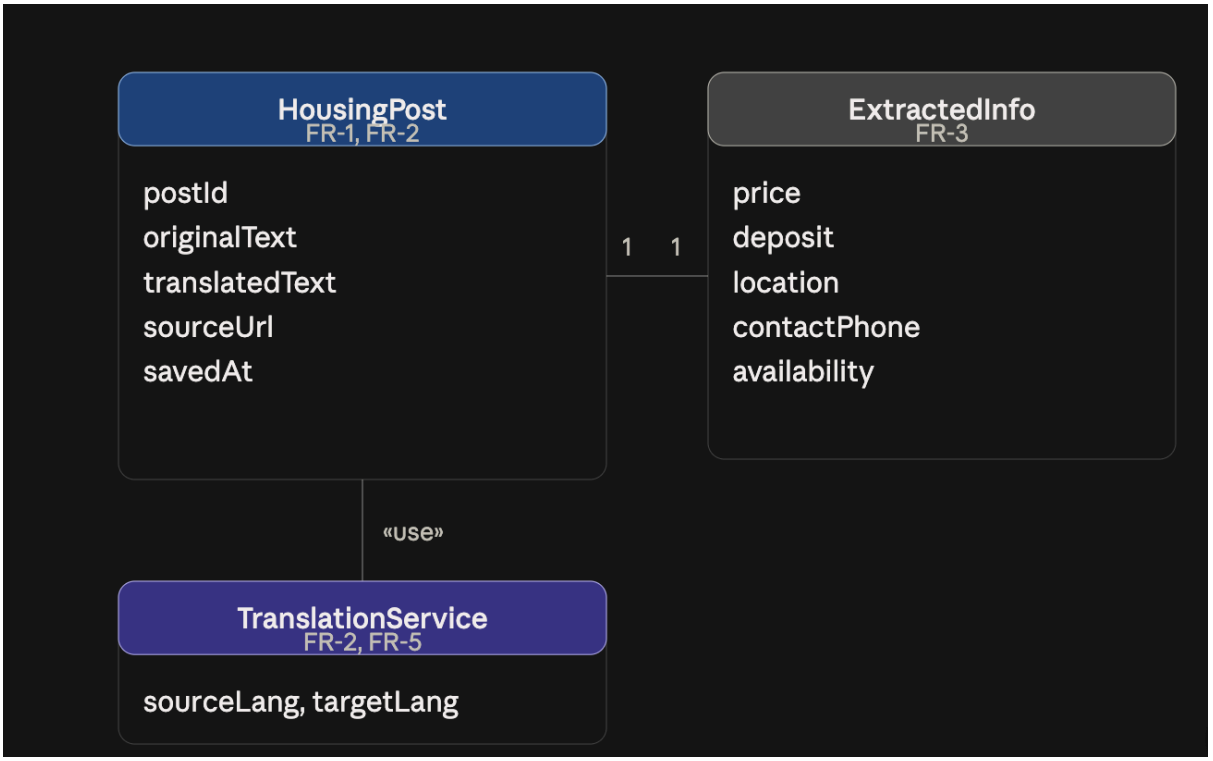
5.1 Use-case list

Use case	Actor	Related FR	Brief success story
UC-1process housing listing	International Student	FR-1	Student pastes a Thai housing post and BaanRight accepts it for processing.
UC-2 translate housing listing	International Student	FR-2	Student receives an English translation of the Thai housing post.
UC-3 Extract and Save housing information	International Student	FR-3	BaanRight extracts key information such as price, deposit, phone number, location, and

Use case	Actor	Related FR	Brief success story
			availability,and saves the information for later review.
UC-4 Generate landlord reply	International Student	FR-4	Student selects a common question and receives a Thai message to copy.
UC-5 Translate selected webpage text	International Student	FR-5	Student selects Thai text on a webpage and translates it using BaanRight.

5.2 Use-case diagram





6. Out of scope

1. Automatically searching Facebook groups.
2. Automatically scraping Facebook posts.
3. Booking or reserving accommodation.
4. Making rental payments.
5. Automatically contacting landlords.
6. Scam or red-flag detection.
7. Guaranteeing that a housing listing is legitimate.
8. Guaranteeing the accuracy of price or availability.
9. Providing legal advice about rental contracts.
10. Determining whether a rental contract is fair or unfair.
11. Managing rental contracts.
12. Shared housing database between students.
13. User login/account system.
14. Building a complete housing marketplace.
15. Screenshot OCR in the current MVP.

7. Traceability matrix (Golden Thread)

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
Thai housing posts are difficult for	FR-2 Translate housing listing	Translation component using MyMemory API	Translate thai to English	Submit Thai housing text and verify an English translation is displayed

Problem / pain	Requirement (FR)	Solution / design	Feature built	Test
international students to understand.				
Housing posts are scattered and difficult to find again.	FR-3 Extract and Save key information	Chrome local storage	saved listings	Save a listing and reopen the saved-listings section
International students may struggle to communicate with Thai landlords.	FR-4 Generate Thai reply	Fixed Thai message templates	Five Common questions for landlord	Select a question and verify Thai message is generated and can be copied
Important details are buried inside long posts.	FR-3 Extract and Save Key information	Rule-based information extraction	Price, deposit, phone, location, availability fields	Test against sample posts with known values

8. AI usage log

If this table is empty but the prose looks generated, the milestone is capped at 50% — log every prompt that drafted a story or paragraph, honestly.

Date	Tool / model	What we asked	What AI produced	What we changed / verified	What we learned
27 Aug 2026	Claude	Help develop the Baanright chrome extension for translating thai housing listing and extract housing information	Generated code and implementation suggestions for translation, extraction, browser storage, and extension features.	Team tested the extension manually and changed implementation where the generated code did not match our requirements.	AI-generated code still needs to be tested and understood by the developers.
27 Aug 2026	Chatgpt	Help organize the project's SRS and convert the housing problem into functional requirements.and Help identify measurable non-functional requirements.	Suggested performance, usability, reliability, privacy, and accuracy requirements.	Team will test the proposed targets and adjust them based on realistic prototype performance.	NFRs should be measurable rather than vague words such as “fast” or “easy.”

9. Self-check

- ☑ Every FR is a user story with a benefit — all 5 FRs in §3 end in a “so that” clause tied to a real pain.
- ☑ Every FR has acceptance criteria you could actually test — 2 AC per FR, each a number, a state, or a yes/no.
- ☑ No FR is vague — no bare “fast”, “easy”, or “user-friendly” anywhere in §3.
- ☑ NFRs exist, including security / privacy — NFR-5 is a required access-control check.
- ☑ Every FR traces back to a stated problem — see “Pain this traces to” under each FR, and §7.
- ☑ Scope and out-of-scope are stated — §2.1 and §6.

- ☑ Use-case list matches the FR list — 5 use cases, one per FR (diagram added Week 4).
- ☑ AI log is honest — §8 shows exactly what was asked, drafted, and hand-corrected.

Course rule: you may use AI to draft. You must log it, verify it against a real stakeholder, and be able to explain every FR in the oral exam. Work you cannot explain earns no credit.